

# Concept Note: Safely Managed Sanitation Hackathon

## Team Name

SaniGuard Alliance

## Project Title

**SaniGuard Hub: Climate-Resilient Smart Sanitation and Waste-to-Energy Retrofit System for Northern Ghana**

---

## 1. Executive Summary

SaniGuard Alliance proposes a climate-resilient sanitation and waste-to-energy solution designed to improve sanitation services in flood-prone and water-scarce communities across Northern Ghana.

SaniGuard Hub is a decentralized sanitation retrofit and monitoring system that upgrades existing sanitation infrastructure using smart monitoring tools, low-cost climate-resilient sanitation technologies, and waste-to-energy recovery systems.

Unlike expensive centralized sanitation systems, SaniGuard focuses on retrofitting and strengthening existing public toilets, waste collection systems, and sanitation facilities to improve resilience during flooding and drought conditions.

The system combines:

- Smart sanitation monitoring,
- Low-cost retrofit sanitation upgrades,
- Waste collection and sorting,
- Anaerobic biodigesters,
- Solar-powered monitoring systems,
- GIS sanitation mapping,
- SMS/USSD community reporting.

Organic and sanitation waste collected from households, public toilets, markets, slaughterhouses, schools, and farms will be processed into:

- clean biogas for cooking,

- organic fertilizer for agriculture,
- eco-friendly briquettes,
- recyclable materials.

The project addresses critical sanitation failures caused by:

- flooding,
- drought,
- weak waste infrastructure,
- open dumping,
- poor sanitation monitoring.

SaniGuard Hub promotes:

- climate-resilient sanitation,
- cleaner communities,
- improved public health,
- renewable energy generation,
- youth employment,
- community-based sanitation management.

---

## 2. Problem Statement

Northern Ghana continues to face major sanitation and waste management challenges, especially during periods of flooding and drought.

Many communities experience:

- overflowing public toilets,
- blocked drainage systems,
- open dumping of waste,
- poor waste collection services,
- contamination of water sources,
- sanitation-related diseases.

Flooding often damages sanitation infrastructure and spreads untreated waste into surrounding communities and water bodies. During drought periods, limited water availability weakens hygiene and sanitation services.

At the same time, large amounts of organic waste from markets, farms, households, and slaughterhouses are left unmanaged, contributing to environmental pollution and methane emissions.

Most sanitation systems in vulnerable communities are poorly monitored and lack reliable sanitation data for emergency response and maintenance planning.

SaniGuard Hub addresses these challenges by upgrading existing sanitation infrastructure with smart climate-resilient retrofit systems and converting waste into valuable energy and agricultural products.

---

## **3. Technical Design and Climate Resilience**

### **3.1 Core System Design**

SaniGuard Hub operates through four integrated systems:

#### **A. Retrofit Sanitation Upgrades**

The project improves existing sanitation infrastructure instead of replacing it completely.

Retrofit upgrades include:

- raised toilet platforms,
- reinforced containment systems,
- sealed waste chambers,
- smart overflow monitoring sensors,
- leak-proof waste transfer systems.

This approach reduces costs while improving sanitation resilience in flood-prone communities.

#### **B. Smart Sanitation Monitoring**

Low-cost IoT sensors monitor:

- toilet fill levels,
- waste tank capacity,
- sanitation overflow risks,
- gas production levels.

The system sends alerts through:

- mobile dashboards,
- SMS notifications,
- USSD reporting tools.

GIS mapping tools identify:

- flood-prone sanitation zones,
- illegal dumping areas,
- waste collection gaps,
- sanitation risk hotspots.

This improves sanitation response, maintenance planning, and emergency coordination.

### **C. Waste-to-Energy Processing**

Organic waste collected from:

- households,
- markets,
- public toilets,
- slaughterhouses,
- farms,

is processed through anaerobic biodigesters.

The system produces:

- clean biogas for cooking,
- organic fertilizer for agriculture.

The biogas is stored in flexible low-pressure storage bags and distributed through community refill stations.

### **D. Recycling and Briquette Production**

Plastic and rubber waste are recycled into reusable products.

Dry combustible waste materials are converted into eco-friendly briquettes for cooking fuel.

---

## **3.2 Flood Adaptation Measures**

SaniGuard Hub is specifically designed for flood-prone communities.

Key flood adaptation measures include:

- elevated sanitation platforms,
- raised biodigesters,

- sealed containment systems,
- smart overflow sensors,
- flood-risk GIS mapping,
- emergency sanitation response support.

These measures reduce contamination risks during extreme rainfall and flooding.

---

### **3.3 Drought Adaptation Measures**

The project also supports water-scarce communities through:

- low-water sanitation systems,
- dry sanitation options,
- solar-powered monitoring systems,
- minimal-water biodigester operation,
- water-saving hygiene education.

Organic fertilizer generated from the system also supports drought-resilient agriculture.

---

## **4. Infrastructure and Technology**

### **Hardware Components**

- Anaerobic biodigesters
- Flexible biogas storage bags
- IoT monitoring sensors
- Solar power systems
- Waste segregation units
- Briquette production equipment
- Recycling units

### **Software Components**

- Mobile sanitation dashboard
- GIS mapping platform
- SMS/USSD reporting tools
- Data analytics system
- Cloud-based monitoring platform

## Materials

- Reinforced PVC biogas bags
  - Corrosion-resistant piping systems
  - Solar-powered battery systems
  - Locally sourced construction materials
- 

## 5. Data Strategy and Operational Logic

Northern Ghana faces major sanitation and waste management data gaps.

SaniGuard Hub generates sanitation data through:

- IoT sensors,
- GIS mapping,
- mobile reporting systems,
- community sanitation volunteers.

The system continuously tracks:

- toilet fill levels,
- flooding incidents,
- sanitation risks,
- waste collection volumes,
- waste generation patterns.

This data supports:

- sanitation planning,
  - emergency response coordination,
  - maintenance scheduling,
  - climate-risk monitoring.
- 

## 6. Pilot Implementation Plan

### Pilot Scope

The initial pilot phase will target:

- 100 households,
- 2 public toilet facilities,
- 1 market area.

## **Expected Outcomes**

The pilot system is expected to:

- process approximately 1 ton of organic waste weekly,
- generate clean cooking biogas,
- produce organic fertilizer for farmers,
- reduce open dumping,
- improve sanitation monitoring.

## **Pilot Prototype Focus During Bootcamp**

During the hackathon bootcamp, the team will prototype:

- a smart sanitation overflow sensor,
- a GIS sanitation monitoring dashboard,
- a small-scale biodigester monitoring system,
- a sanitation retrofit system model.

---

# **7. Maintenance and Sustainability**

SaniGuard Hub will operate through a community-managed sanitation model.

## **Key Stakeholders**

- Community sanitation committees
- Local Assemblies
- Youth sanitation entrepreneurs
- Trained technicians
- SaniGuard operational team

## **Maintenance Strategy**

- routine inspection schedules,
- local technician training,
- emergency repair support,
- community maintenance funds,

- solar-powered backup systems.

Following floods or climate shocks, trained local response teams will inspect and restore damaged systems.

---

## 8. Social Equity and Inclusion

SaniGuard prioritizes:

- low-income households,
- women and girls,
- rural communities,
- flood-prone settlements,
- schools and public institutions.

The project promotes inclusion through:

- affordable sanitation services,
- community ownership,
- local employment opportunities,
- youth-led sanitation operations,
- use of local materials and labor.

The modular retrofit model allows District Assemblies to expand gradually at lower cost.

---

## 9. Revenue and Sustainability Model

SaniGuard Hub uses a circular economy model.

Revenue streams include:

- biogas sales,
- organic fertilizer sales,
- briquette sales,
- recycling products,
- sanitation service fees.

These revenue streams support:

- operational sustainability,



- maintenance costs,
  - community expansion,
  - long-term climate resilience.
- 

## 10. Expected Impact

The pilot project is expected to:

- improve sanitation access for over 500 residents,
- reduce unsafe waste disposal,
- provide cleaner cooking fuel alternatives,
- support flood resilience,
- reduce environmental pollution,
- improve sanitation monitoring,
- create youth employment opportunities.

Long-term implementation can support healthier and cleaner communities across Northern Ghana.

---

## 11. Sustainable Development Goals (SDGs)

SaniGuard Hub contributes to:

- SDG 3: Good Health and Well-being
  - SDG 6: Clean Water and Sanitation
  - SDG 7: Affordable and Clean Energy
  - SDG 11: Sustainable Cities and Communities
  - SDG 13: Climate Action
- 

## 12. Risks and Mitigation

### Major Risk

Financial and technical sustainability during early deployment.

### Mitigation Measures

- Start with a small pilot community,
  - Use low-cost locally sourced materials,
  - Partner with Local Assemblies and NGOs,
  - Train local youth technicians,
  - Generate revenue from biogas and recycling,
  - Expand gradually after pilot validation.
- 

## 13. Conclusion

SaniGuard Alliance believes sanitation systems should not only manage waste, but also create clean energy, economic opportunity, environmental protection, and climate resilience.

By upgrading existing sanitation systems with climate-smart retrofit technologies, smart monitoring tools, and waste-to-energy processing, SaniGuard Hub provides an affordable and scalable sanitation solution for vulnerable communities across Northern Ghana.

The project transforms sanitation challenges into opportunities for cleaner communities, renewable energy generation, climate adaptation, and sustainable livelihoods.